

Lab 2: Empirical Risk and Simple Linear Regression

EECS 245, Fall 2026 at the University of Michigan

due by the end of your lab section on Wednesday, September 9th, 2026

Name: _____

username: _____

Each lab worksheet will contain several activities, some of which will involve writing code and others that will involve writing math on paper. To receive credit for a lab, you must complete all required activities and show your lab TA by the end of the lab section.

While you must get checked off by your lab TA **individually**, we encourage you to form groups with 1-2 other students to complete the activities together.

Recap: The Modeling Recipe

In [Chapter 1.3](#), we introduced the three-step modeling recipe for finding optimal model parameters, which ultimately helps us make the best possible predictions.

1. Choose a model.

$$\underbrace{h(x_i) = w}_{\text{constant model}} \quad \underbrace{h(x_i) = w_0 + w_1 x_i}_{\text{simple linear regression model}}$$

2. Choose a loss function.

$$\underbrace{L_{\text{sq}}(y_i, h(x_i)) = (y_i - h(x_i))^2}_{\text{squared loss}} \quad \underbrace{L_{\text{abs}}(y_i, h(x_i)) = |y_i - h(x_i)|}_{\text{absolute loss}}$$

3. Minimize *average loss* (also called *empirical risk*) to find optimal model parameters.

- Constant model, squared loss: $R_{\text{sq}}(w) = \frac{1}{n} \sum_{i=1}^n (y_i - w)^2 \implies w^* = \bar{y}$
- Constant model, absolute loss: $R_{\text{abs}}(w) = \frac{1}{n} \sum_{i=1}^n |y_i - w| \implies w^* = \text{Median}(y_1, y_2, \dots, y_n)$
- Simple linear regression model, squared loss:

$$R_{\text{sq}}(w_0, w_1) = \frac{1}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x_i))^2 \implies w_1^* = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}, \quad w_0^* = \bar{y} - w_1^* \bar{x}$$

Activity 1: Relative Squared Loss

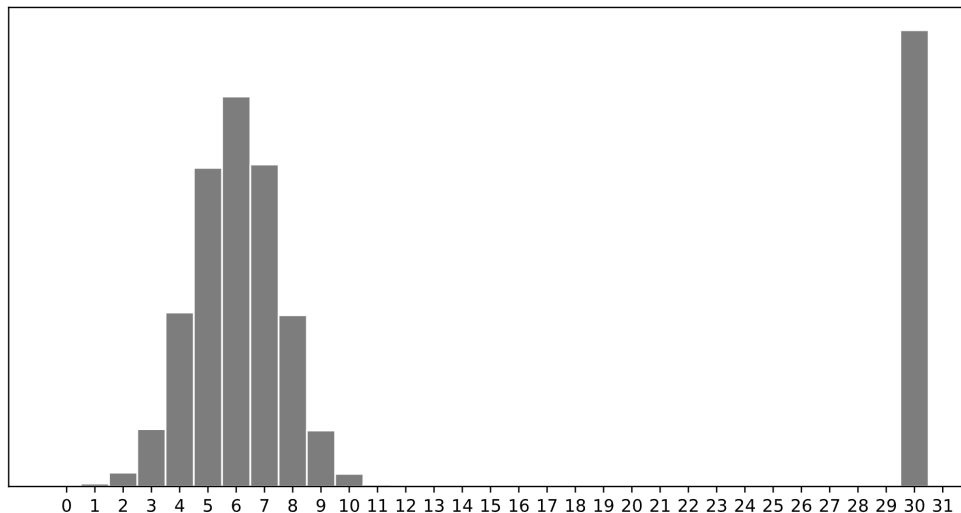
Suppose we'd like to find the optimal parameter, w^* , for the constant model $h(x_i) = w$. To do so, we use the following loss function, called the **relative squared loss**:

$$L_{\text{rsq}}(y_i, h(x_i)) = \frac{(y_i - h(x_i))^2}{y_i}$$

What value of w minimizes the average loss (i.e. empirical risk) when using the relative squared loss function – that is, what is w^* ? Your answer should only be in terms of the variables $n, y_1, y_2, \dots, y_n$, and any constants.

Activity 2: Rapid Fire

Consider a dataset of n **integers**, $y_1, y_2, \dots, y_n$, whose histogram is given below:



a) Which of the following is closest to the constant prediction w^* that minimizes:

$$\frac{1}{n} \sum_{i=1}^n \begin{cases} 0 & y_i = w \\ 1 & y_i \neq w \end{cases}$$

- ☐ 1 ☐ 5 ☐ 6 ☐ 7 ☐ 11 ☐ 15 ☐ 30

b) Which of the following is closest to the constant prediction w^* that minimizes:

$$\frac{1}{n} \sum_{i=1}^n |y_i - w|$$

- ☐ 1 ☐ 5 ☐ 6 ☐ 7 ☐ 11 ☐ 15 ☐ 30

c) Which of the following is closest to the constant prediction w^* that minimizes:

$$\frac{1}{n} \sum_{i=1}^n (y_i - w)^2$$

- ☐ 1 ☐ 5 ☐ 6 ☐ 7 ☐ 11 ☐ 15 ☐ 30

d) Which of the following is closest to the constant prediction w^* that minimizes:

$$\lim_{p \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n |y_i - w|^p$$

Hint: Think about the effect of outliers.

- ☐ 1 ☐ 5 ☐ 6 ☐ 7 ☐ 11 ☐ 15 ☐ 30

Activity 3: Slope of Mean Absolute Error

Consider a dataset of 8 points, $y_1, y_2, \dots, y_8$ that are in sorted order, i.e. $y_1 < y_2 < \dots < y_8$.

Recall that mean absolute error, $R_{\text{abs}}(w)$, for the constant model $h(x_i) = w$ is defined as:

$$R_{\text{abs}}(w) = \frac{1}{n} \sum_{i=1}^n |y_i - w|$$

This is a piecewise linear function that changes slope at each data point. The slope of $R_{\text{abs}}(w)$ at any w that is not a data point is:

$$\frac{d}{dw} R_{\text{abs}}(w) = \frac{\text{number left of } w - \text{number right of } w}{n}$$

Suppose that $y_4 = 10$, $y_5 = 14$, $y_6 = 22$, and $R_{\text{abs}}(11) = 9$. What is $R_{\text{abs}}(22)$?

Hint: You don't have all 8 of the y -values, so you can't find $R_{\text{abs}}(22)$ just by plugging in numbers into the formula for $R_{\text{abs}}(w)$. Instead, think about how to use the slope formula.

Activity 4: Programming

Complete the tasks in the `lab02.ipynb` notebook.

There are two ways to access the supplemental Jupyter Notebook:

- **Option 1 (preferred):** Set up a Jupyter Notebook environment locally, use `git` to clone our [course repository](#), and open `labs/lab02/lab02.ipynb`. For instructions on how to do this, see the [Environment Setup](#) page of the course website.
- **Option 2:** Click [here](#) to open `lab02.ipynb` on DataHub. Before doing so, read the instructions on the [Environment Setup](#) page on how to use the DataHub.

To receive credit for Activity 4, you'll need to show your lab TA that all test cases have passed.

Activity 5: What Do You Mean?

Suppose we want to fit a simple linear model (using squared loss) that predicts the number of ingredients in a product given its price. We're given that:

- The average cost of a product in our dataset is \$40, i.e. $\bar{x} = 40$
- The average number of ingredients in a product in our dataset is 15, i.e. $\bar{y} = 15$

The intercept and slope of the regression line are $w_0^* = 11$ and $w_1^* = \frac{1}{10}$, respectively.

- a) Suppose Victors' Veil (a skincare product) costs \$40 and has 11 ingredients. What is the squared loss of our model's predicted number of ingredients for Victors' Veil?

- b) Is it possible to answer part a) above **just** by knowing $\bar{x}$ and $\bar{y}$, i.e. **without** knowing the values of w_0^* and w_1^* ? Once you select an answer, explain it to your peers.

Hint: If you're stuck, look through the headings of the sections of [Chapter 2.3](#).

☐ Yes, it's possible ☐ No, it's not possible

Activity 6: The Meaning of Mean Squared Error

Suppose we'd like to predict the number of minutes a delivery will take, y , as a function of distance, x . To do so, we look to our dataset of n deliveries, $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, and fit two simple linear models:

- $F(x_i) = a_0 + a_1 x_i$, where:

$$a_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}, \quad a_0 = \bar{y} - a_1 \bar{x}$$

Here, $\bar{x}$ and $\bar{y}$ are the respective means of the x -values and y -values.

- $G(x_i) = b_0 + b_1 x_i$, where b_0 and b_1 are chosen such that $G(x_i) = b_0 + b_1 x_i$ minimizes **mean absolute error** on the dataset. Assume that no other line minimizes mean absolute error on the dataset, i.e. that the values of b_0 and b_1 are unique. Assume also that F and G are distinct lines.

a) Fill in the

$$\sum_{i=1}^n (y_i - F(x_i))^2 \quad \text{} \quad \sum_{i=1}^n (y_i - G(x_i))^2$$

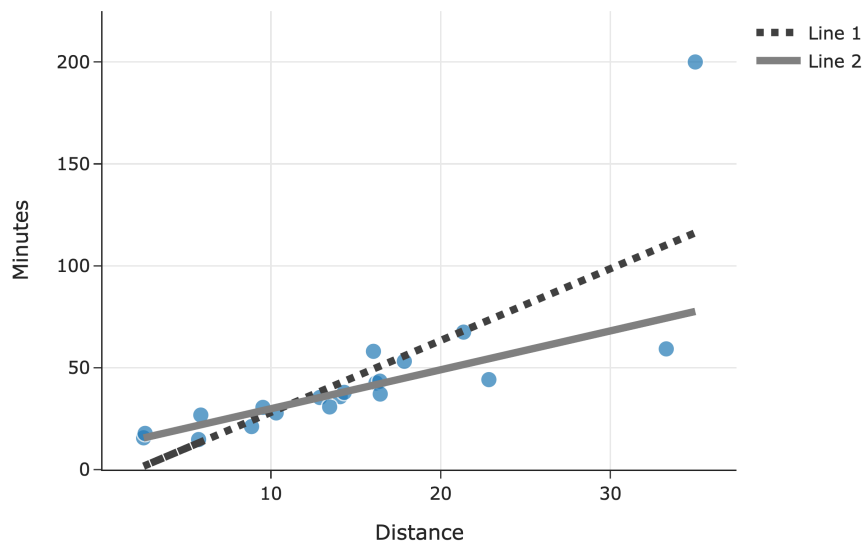
☐ $>$ ☐ $\geq$ ☐ $=$ ☐ $\leq$ ☐ $<$ ☐ Impossible to tell

b) Fill in the

$$\left(\sum_{i=1}^n |y_i - F(x_i)| \right)^2 \quad \text{} \quad \left(\sum_{i=1}^n |y_i - G(x_i)| \right)^2$$

☐ $>$ ☐ $\geq$ ☐ $=$ ☐ $\leq$ ☐ $<$ ☐ Impossible to tell

c) Below, we've drawn the lines for both F and G along with a scatter plot for the original n deliveries:



Which line corresponds to F ?

☐ Line 1

☐ Line 2

The following are extra practice. Don't feel pressured to answer all of these problems in lab, but make sure to attempt them at some point.

Activity 7: Relative Squared Loss, Continued

Recall the formula for **relative squared loss** from Activity 1:

$$L_{\text{rsq}}(y_i, h(x_i)) = \frac{(y_i - h(x_i))^2}{y_i}$$

- a) Let $C(y_1, y_2, \dots, y_n)$ be your minimizer w^* from Activity 1. That is, for a particular dataset $y_1, y_2, \dots, y_n$, $C(y_1, y_2, \dots, y_n)$ is the value of w that minimizes empirical risk for relative squared loss on that dataset.

What is the value of $\lim_{y_4 \rightarrow \infty} C(1, 3, 5, y_4)$ in terms of $C(1, 3, 5)$? Your answer should involve the function C and/or one or more constants.

Hint: To notice the pattern, evaluate $C(1, 3, 5, 100)$, $C(1, 3, 5, 10000)$, and $C(1, 3, 5, 1000000)$.

- b) What is the value of $\lim_{y_4 \rightarrow 0} C(1, 3, 5, y_4)$? Again, your answer should involve the function C and/or one or more constants.

- c) Based on the results of the previous two parts, when is the prediction $C(y_1, y_2, \dots, y_n)$ robust to outliers? When is it not robust to outliers?

Activity 8: A Refresher

Consider a dataset of $y_1, y_2, \dots, y_n$, all of which are **positive**. We want to fit a constant model, $h(x_i) = w$, to the data.

Let w_p^* be the optimal constant prediction that minimizes average degree- p loss, $R_p(w)$, defined below:

$$R_p(w) = \frac{1}{n} \sum_{i=1}^n |y_i - w|^p$$

For example, w_2^* is the optimal constant prediction that minimizes $R_2(w) = \frac{1}{n} \sum_{i=1}^n |y_i - w|^2$

a) In each of the parts below, determine the value of the quantity provided. By “the data”, we are referring to $y_1, y_2, \dots, y_n$. The answer choices are as follows; **select one item in each row**.

- A: The standard deviation of the data
- B: The variance of the data
- C: The mean of the data
- D: The median of the data
- E: The midrange of the data, $\frac{y_{\min} + y_{\max}}{2}$
- F: The mode of the data
- G: None of these

	A	B	C	D	E	F	G
(i) h_0^*	☐	☐	☐	☐	☐	☐	☐
(ii) h_1^*	☐	☐	☐	☐	☐	☐	☐
(iii) $R_1(h_1^*)$	☐	☐	☐	☐	☐	☐	☐
(iv) h_2^*	☐	☐	☐	☐	☐	☐	☐
(v) $R_2(h_2^*)$	☐	☐	☐	☐	☐	☐	☐

b) Now, suppose we want to find the optimal constant prediction, h_U^* , using the “Ultra” loss function, defined below:

$$L_U(y_i, w) = y_i(y_i - w)^2$$

To find h_U^* , we minimize $R_U(w)$, the average Ultra loss. How does h_U^* compare to the mean of the data, M ?

- ☐ $h_U^* > M$
 ☐ $h_U^* \geq M$
 ☐ $h_U^* = M$
 ☐ $h_U^* \leq M$
 ☐ $h_U^* < M$

- c) Finally, to find the optimal constant prediction, we will instead minimize **regularized** average Ultra loss, $R_\lambda(w)$, defined below:

$$R_\lambda(w) = \left(\frac{1}{n} \sum_{i=1}^n y_i (y_i - w)^2 \right) + \lambda w^2$$

Here, assume $\lambda > 0$ is some positive constant. (We will cover regularization in more detail later in the term.)

Find w^* , the constant prediction that minimizes $R_\lambda(w)$. Give your answer as an expression in terms of the y_i 's, n , and/or λ .